

Mobile App User Interaction Analysis

Exploratory Data Analysis & Statistical Hypothesis Testing

Dataset: `mobile_app_interactions_csv.csv`

Tools: Python · Pandas · Matplotlib · Seaborn · SciPy

May 6, 2026

Contents

1	Introduction	2
2	Data Loading & Initial Inspection	2
3	Data Cleaning & Preparation	3
3.1	Removing Unnecessary Columns	3
3.2	Handling Missing Values	3
3.3	Standardising Categorical Columns	4
3.4	Session Duration: Outlier Removal	6
4	Exploratory Data Analysis	6
4.1	Session Duration Distribution	6
4.2	Session Duration by Event Type	8
4.3	Session Duration by Subscription & Push Status	9
5	Hypothesis Testing	9
5.1	Test 1 — Subscription Status vs. Session Duration	9
5.2	Test 2 — Push Notifications vs. Session Duration	9
6	Findings & Conclusion	10
6.1	Results Summary	10
6.2	Key Takeaways	10

Introduction

This report analyses the **Mobile App User Interaction** dataset, which captures how users engage with a mobile application — recording session durations, interaction types (clicks, scrolls, swipes, etc.), device OS, subscription status, and push notification preferences.

Objectives:

- Clean and prepare raw data for analysis.
- Uncover patterns in user behaviour through EDA.
- Test whether subscription status and push notifications significantly affect session duration.

Data Loading & Initial Inspection

The dataset was loaded and inspected using standard Pandas functions to understand its structure, column types, and completeness.

```
df = pd.read_csv("mobile_app_interactions_csv.csv")
df.head()
df.info()
df.describe(include="all")
```

	user_id	session_id	device_os	event_type	app_version	session_duration_sec	is_subscribed	user_age	push_enabled	Unnamed: 9
0	13457	S-852694	Android	double_tap	2.6.8	3311	TRUE	51	FALSE	NaN
1	95977	S-766538	Android	pinch	3.3.6	3283	FALSE	40	FALSE	NaN
2	52758	S-228244	iOS	zoom	5.4.5	2870	TRUE	73	TRUE	NaN
3	53777	S-633216	iOS	long_press	4.7.1	332	TRUE	47	FALSE	NaN
4	6302	S-508537	Android	view	2.0.8	1694	TRUE	32	FALSE	NaN

Figure 1: First five rows of the dataset

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   user_id               99659 non-null   str
1   session_id           99680 non-null   str
2   device_os            99686 non-null   str
3   event_type           99650 non-null   str
4   app_version          99670 non-null   str
5   session_duration_sec 99680 non-null   str
6   is_subscribed        99660 non-null   str
7   user_age             99649 non-null   str
8   push_enabled         99662 non-null   str
9   Unnamed: 9           0 non-null      float64
dtypes: float64(1), str(9)
memory usage: 7.6 MB
```

Figure 2: Column types and non-null counts

Data Cleaning & Preparation

The raw dataset had several quality issues that were resolved before analysis.

Removing Unnecessary Columns

An artefact column `Unnamed: 9` (a CSV export side-effect) was dropped.

```
df = df.drop(columns=['Unnamed: 9'])
```

Handling Missing Values

- **Critical rows** (`user_id`, `session_id`, `session_duration_sec`) — rows dropped entirely.
- **Categorical columns** (`device_os`, `event_type`, `app_version`) — filled with "Unknown".
- `user_age` — converted to numeric; missing values imputed with the **median**.
- **Boolean flags** (`is_subscribed`, `push_enabled`) — nulls filled with `False`.

```
df = df.dropna(subset=["user_id", "session_id", "session_duration_sec"])
df["device_os"] = df["device_os"].fillna("Unknown")
df["event_type"] = df["event_type"].fillna("Unknown")
df["user_age"] = pd.to_numeric(df["user_age"], errors="coerce")
df["user_Age"] = df["user_age"].fillna(df["user_age"].median())
df["is_subscribed"] = df["is_subscribed"].fillna(False)
df["push_enabled"] = df["push_enabled"].fillna(False)
```

```
: df.isnull().sum()
```

```
: user_id                                0
   session_id                            0
   device_os                             0
   event_type                            0
   app_version                           0
   session_duration_sec                  0
   is_subscribed                         0
   user_age                              0
   push_enabled                          0
   dtype: int64
```

Figure 3: Zero null values confirmed after cleaning

Standardising Categorical Columns

Both `device_os` and `event_type` contained misspelled and corrupted labels. These were corrected using replacement maps and filtered to valid values only.

```
# Device OS
df["device_os"] = df["device_os"].str.lower().str.strip()
df["device_os"] = df["device_os"].replace({"android":"android", "andromd":"android",
,
                                           "ioo":"ios", "io6":"ios"})
```

```

df["device_os"] = df["device_os"].apply(
    lambda x: x if x in ["android","ios"] else "unknown")

# Event Type
df["event_type"] = df["event_type"].str.lower().str.strip()
df["event_type"] = df["event_type"].replace({"koom":"zoom", "lonx_press":"
    long_press",
                                           "shar4":"share", "hxu8se":"tap",
                                           "mu8svw":"tap"})
valid_et = ["click","share","swipe","scroll","view","zoom","long_press","tap"]
df["event_type"] = df["event_type"].apply(
    lambda x: x if x in valid_et else "unknown")
df_clean = df[df["event_type"] != "unknown"]

```

```

: df_clean[["device_os","event_type"]].value_counts()

```

```

: device_os  event_type
  android    click      5649
          share      5648
          long_press  5599
          zoom      5557
          scroll      5539
          view      5524
          swipe      5507
  ios       swipe      4240
          share      4238
          click      4231
          view      4206
          scroll      4163
          zoom      4122
          long_press  4103
  unknown   scroll      111
          swipe      102
          zoom      100
          click      97
          share      82
          long_press  78
          view      74
  ios       tap        2
Name: count, dtype: int64

```

Figure 4: Distribution of device OS and event types after standardisation

Session Duration: Outlier Removal

Session duration was coerced to numeric, missing values imputed with the median, and extreme values ($\geq 10,000$ seconds) removed.

```
df_clean["session_duration_sec"] = pd.to_numeric(
    df_clean["session_duration_sec"], errors="coerce")
df_clean["session_duration_sec"].fillna(
    df_clean["session_duration_sec"].median(), inplace=True)
df_clean = df_clean[df_clean["session_duration_sec"] < 10000]
```

Exploratory Data Analysis

Session Duration Distribution

```
sns.histplot(df_clean['session_duration_sec'])
plt.title("Session Duration Distribution")
plt.show()

sns.boxplot(x=df_clean['session_duration_sec'])
plt.title("Session Duration Boxplot")
plt.show()
```

```
sns.histplot(df_clean['session_duration_sec'])  
plt.title("Session Duration Distribution")  
plt.xlabel("Session Duration")  
plt.show()
```

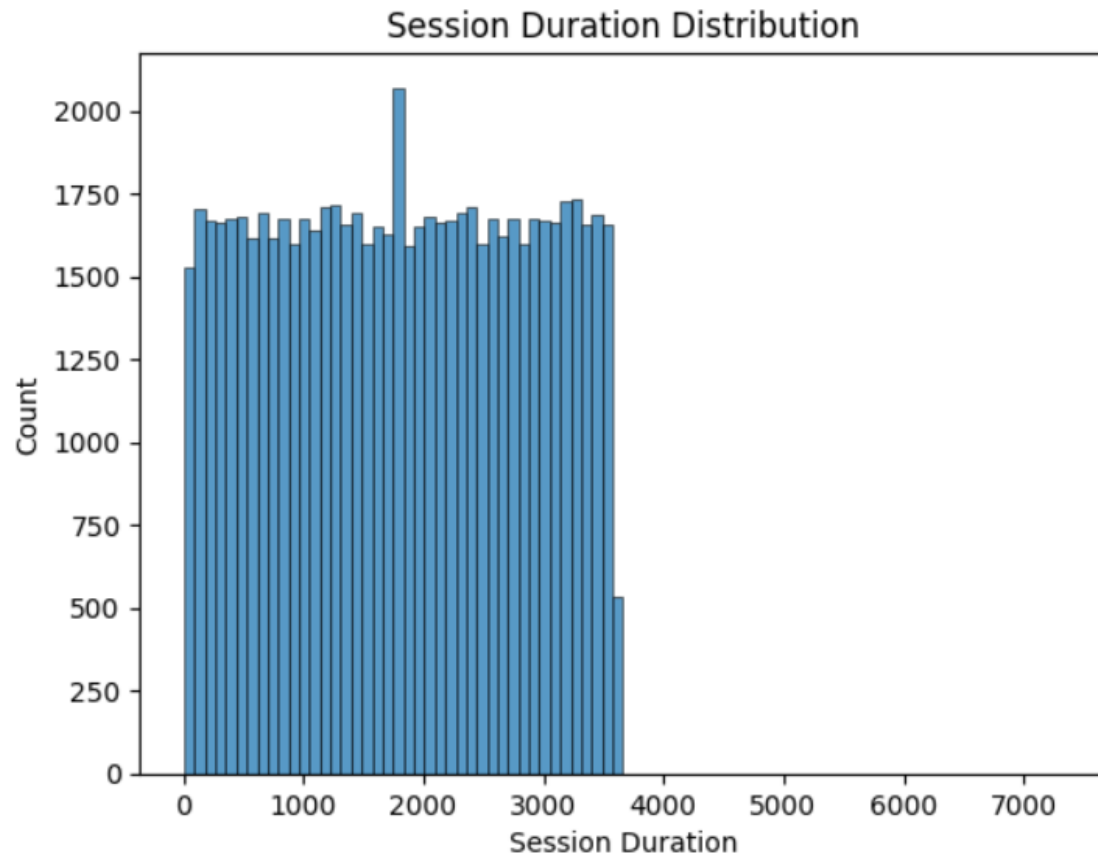


Figure 5: Distribution of session duration across all users


```
sns.boxplot(x=df_clean['session_duration_sec'])  
plt.title("Session Duration Boxplot")  
plt.show()
```

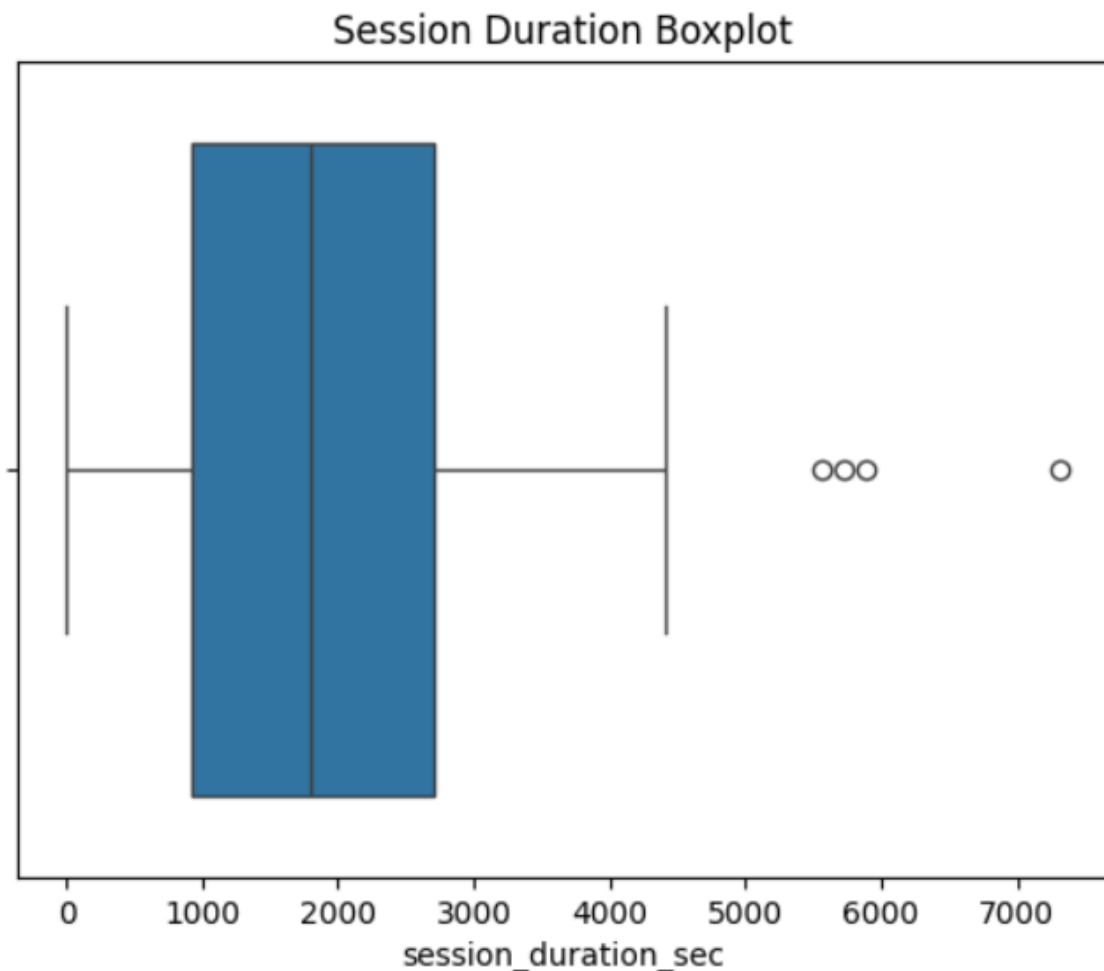


Figure 6: Median $\approx$ 1800 sec ($\sim$ 30 min); outliers indicate power users

Key Insight

Session duration is approximately uniformly distributed with high variability. The median is around 30 minutes, and high-value outliers represent a segment of highly engaged “power users”.

Session Duration by Event Type

```
df_clean.groupby('event_type')['session_duration_sec'].mean().plot(kind='bar')  
plt.title("Average Session Duration by Event Type")  
plt.xlabel("Event Type")  
plt.ylabel("Avg Duration (sec)")  
plt.grid()  
plt.show()
```

Key Insight

Session duration is broadly consistent across event types. Deeper interactions such as *scroll* and *zoom* show slightly longer sessions, while *tap* shows a lower average — likely due to limited data for that event.

Session Duration by Subscription & Push Status

```
df_clean.groupby('is_subscribed')['session_duration_sec'].mean()
df_clean.groupby('push_enabled')['session_duration_sec'].mean()
```

Key Insight

Both subscribed and non-subscribed users, and users with push on or off, show nearly identical mean session durations — these features do not independently drive engagement.

Hypothesis Testing

Two independent two-sample *t*-tests were conducted at significance level $\alpha = 0.05$.

Test 1 — Subscription Status vs. Session Duration

$$H_0 : \mu_{\text{subscribed}} = \mu_{\text{not subscribed}} \quad H_1 : \mu_{\text{subscribed}} \neq \mu_{\text{not subscribed}}$$

```
from scipy.stats import ttest_ind
subscribed = df_clean[df_clean['is_subscribed'] == True]['session_duration_sec']
not_subscribed = df_clean[df_clean['is_subscribed'] == False]['session_duration_sec']
ttest_ind(subscribed, not_subscribed)
```

```
from scipy.stats import ttest_ind

subscribed = df_clean[df_clean['is_subscribed'] == True]['session_duration_sec']
not_subscribed = df_clean[df_clean['is_subscribed'] == False]['session_duration_sec']

ttest_ind(subscribed, not_subscribed)

TtestResult(statistic=np.float64(-0.6987615369910434), pvalue=np.float64(0.48470342567243135), df=np.float64(68969.0))
```

Figure 7: *t*-test output for subscription vs. session duration

Result

$$t = -0.699, \quad p = 0.485 > 0.05 \quad \Rightarrow \text{Fail to reject } H_0$$

Subscription status has **no statistically significant effect** on session duration.

Test 2 — Push Notifications vs. Session Duration

$$H_0 : \mu_{\text{push=on}} = \mu_{\text{push=off}} \quad H_1 : \mu_{\text{push=on}} \neq \mu_{\text{push=off}}$$

```
push_true = df_clean[df_clean['push_enabled'] == True]['session_duration_sec']
push_false = df_clean[df_clean['push_enabled'] == False]['session_duration_sec']
ttest_ind(push_true, push_false)
```

```

from scipy.stats import ttest_ind

push_true = df_clean[df_clean['push_enabled'] == True]['session_duration_sec']
push_false = df_clean[df_clean['push_enabled'] == False]['session_duration_sec']

ttest_ind(push_true, push_false)

TtestResult(statistic=np.float64(-0.08261124192158777), pvalue=np.float64(0.9341609018099056), df=np.float64(68969.0))

```

Figure 8: t-test output for push notifications vs. session duration

Result

$t = -0.083$, $p = 0.934 > 0.05 \Rightarrow$ **Fail to reject H_0**

Push notification status has **no statistically significant effect** on session duration.

Findings & Conclusion

Results Summary

Table 1: Hypothesis testing summary

Test	t -stat	p -value	Decision	Outcome
Subscription vs. Duration	-0.699	0.485	Fail to reject H_0	No significant effect
Push Notifications vs. Duration	-0.083	0.934	Fail to reject H_0	No significant effect

Key Takeaways

- **Top interactions:** Click, Share, and Scroll dominate user activity.
- **Device split:** Android is the majority platform, followed by iOS.
- **Session duration:** Averages ≈ 30 minutes; uniformly distributed with high variability.
- **Power users:** High-duration outliers represent a deeply engaged segment worth further study.
- **Feature flags don't drive engagement:** Neither subscription nor push notification status significantly influences session duration.

Final Conclusion

User engagement is not significantly driven by subscription plans or push notification settings. Improving **content quality**, **personalisation**, and the overall **app experience** are likely to be far more effective strategies for increasing session duration.